

1.5 Littlewood's Conjecture

For a real number t , write $\langle t \rangle$ for the distance from t to the nearest integer,

$$\langle t \rangle = \min_{q \in \mathbb{Z}} |t - q|.$$

The theory of continued fractions (which will be described in Chap. 3) shows that for any real number u , there is a sequence (q_n) with $q_n \rightarrow \infty$ such that $q_n \langle q_n u \rangle < 1$ for all $n \geq 1$. Littlewood conjectured the following in the 1930s: for any real numbers u, v ,

$$\liminf_{n \rightarrow \infty} n \langle nu \rangle \langle nv \rangle = 0.$$

Some progress was made on this for restricted classes of numbers u and v by Cassels and Swinnerton-Dyer [50], Pollington and Velani [290], and others, but the problem remains open. In 2003 Einsiedler, Katok and Lindenstrauss [79] used ergodic methods to prove that the set of exceptions to Littlewood's conjecture is extremely small.

Theorem (Einsiedler, Katok & Lindenstrauss). Let

$$\Theta = \left\{ (u, v) \in \mathbb{R}^2 \mid \liminf_{n \rightarrow \infty} n \langle nu \rangle \langle nv \rangle > 0 \right\}.$$

Then the Hausdorff dimension of Θ is zero.

In fact the result in [79] is a little stronger, showing that Θ satisfies a stronger property that implies it has Hausdorff dimension zero. The proof relies on a partial classification of certain invariant measures on $\mathrm{SL}_3(\mathbb{Z}) \backslash \mathrm{SL}_3(\mathbb{R})$. This is part of the theory of *measure rigidity*, and the particular type of phenomenon seen has its origins in work of Furstenberg [100], who showed that the natural action $t \mapsto at \pmod{1}$ of the semi-group generated by two multiplicatively independent natural numbers a_1 and a_2 on $\mathbb{T}$ has, apart from finite sets, no non-trivial closed invariant sets. He asked if this system could have any non-atomic ergodic invariant measures other than Lebesgue measure. Partial results on this and related generalizations led to the formulation of far-reaching conjectures by Margulis [251], by Furstenberg, and by Katok and Spatzier [183, 184]. A special case of these conjectures concerns actions of the group A of positive diagonal matrices in $\mathrm{SL}_k(\mathbb{R})$ for $k \geq 3$ on the space $\mathrm{SL}_k(\mathbb{Z}) \backslash \mathrm{SL}_k(\mathbb{R})$: if μ is an A -invariant ergodic probability measure on this space, is there a closed connected group $L \geq A$ for which μ is the unique L -invariant measure on a single closed L -orbit (that is, is μ *homogeneous*)?

In the work of Einsiedler, Katok and Lindenstrauss the conjecture stated above is proved under the additional hypothesis that the measure μ gives positive entropy to some one-parameter subgroup of A , which leads to the